

Phase 6: Agile Execution Plan

Contains Epics, User Stories, 2-Sprint Plan and a Risk Register along with Success Measures.

Epics

- Epic 1: Data pipeline for consistency scoring.
- Epic 2: Leaderboard Visualization
- Epic 3: Publishing & Sharing

User Stories

- As a race strategist, I want to see which drivers manage tire wear most consistently, so that I can weigh their in race feedback more accurately.
- As a race strategist, I want the ranking to exclude drivers with too few laps on a compound so that the small sample noise doesn't mess up the leaderboard.
- As a viewer of the dashboard, I want to filter the leaderboard by race or compound so that I can compare consistency in different conditions.
- As someone sharing this analysis on social media, I want a clean, presentable table so that the insight is easy to understand without explanation.
- Having an AI chatbot that does not guess but returns answers based on calculated/available data and clearly states if it doesn't know the answer either helps a lot for casual F1 viewers like me who don't have the understanding or knowledge to read complicated analysis or dashboard/charts..

2 Sprint Plan

Sprint 1:

Initially we pulled the race/session data using FastF1 API for three races: Bahrain, Monaco and Monza. We did a basic understanding of the data shape, structure and size. To perform the tire degradation analysis, we exported the driver/compound data with the information like LapTimeSeconds, TireLife etc.

While trying to perform the tire degradation analysis, we came across the problem of cliff laps being too early or unrealistic. This was due to any first lap chaos or any yellow flag coming up which caused the driver to slow down.

So we introduced a minimum lap filter to tackle this issue. By setting a minimum lap filter of 1 lap and a rolling period of 3 laps helps keep the data consistent. We have put the delta offset by around 1.5 seconds now which can be modified.

Sprint 2:

Once the degradation and consistency calculations were checked, we proceeded with building the actual leaderboard people would see. We built a Power BI table using the CV-based consistency score and sorted it so the most consistent drivers show up first. We also renamed the columns so it's easily understandable without needing to explain the math. We added filters for the race and compound so viewers can see the results according to their preferences. Once the table looked clean we exported screenshots to use for the LinkedIn post.

Risk Register

Risk	Likelihood	Impact	Mitigation
Small sample sizes mess up rankings	High	High	Keep a minimum lap count filter
The raw std unfairly punishes the slower compounds	High	Medium	Switch to coefficient of variation instead of raw standard deviation
Only 3 races so far. Sample size limited. Ranking may not generalize.	Medium	Medium	Flagged under Next Roadmap. Scaling project with multi season benchmarking
Casual viewers may misunderstand consistency as pure skill and not consider car or team factors.	Medium	Low	Clear labelling and caveats in the writeup.

Success Measures

- The ranking gives somewhat of a 'sniff test' against known driver reputations.
- Engagement on the LinkedIn posts.
- The pipeline becomes usable for the next roadmap item (multi season benchmarking)